

KŁODZKO, Poland

WMO: 125200

Lat: 50.4369N

Lon: 16.6142E

Elev: 357

StdP: 97.11

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-12.3	-17.4	0.8	-14.8	-14.7	1.1	-11.7	13.8	0.5	12.3	0.9	1.1	20	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.9	20.0	26.9	19.1	25.2	18.2	20.7	27.1	19.7	25.5	18.8	23.9	3.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	14.0	23.5	17.7	13.2	22.2	16.8	12.5	21.3	61.5	27.2	58.0	25.7	54.8	24.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
10.6	9.1	7.8	-18.7	31.9	4.5	1.6	-21.9	33.0	-24.5	33.9	-27.1	34.8	-30.3	36.0		
			-18.9	22.8	4.5	1.1	-22.1	23.6	-24.8	24.2	-27.3	24.8	-30.5	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-2.0	-0.5	3.0	8.1	12.3	15.6	17.7	17.4	13.0	8.6	4.0	-0.4
	DBStd	8.01	5.40	5.20	4.31	4.08	3.51	3.28	2.92	3.06	3.29	4.02	4.23	4.88
	HDD10.0	1576	372	294	219	83	17	1	0	0	9	74	182	323
	HDD18.3	3799	631	527	476	308	188	94	48	54	163	301	429	581
	CDD10.0	887	0	0	2	25	89	169	238	231	98	32	3	0
	CDD18.3	69	0	0	0	0	2	12	28	26	2	0	0	0
	CDH23.3	851	0	0	0	4	30	144	339	301	33	0	0	0
	CDH26.7	196	0	0	0	0	2	25	87	77	4	0	0	0
Wind	WSAvg	2.8	3.5	3.5	3.3	2.8	2.4	2.0	2.1	2.0	2.2	3.0	3.4	3.6
Precipitation	PrecAvg	812	56	48	60	47	79	88	112	83	73	54	55	57
	PrecMax	1007	110	104	146	87	156	167	341	263	142	135	98	105
	PrecMin	567	15	8	19	9	18	29	26	32	20	7	1	21
	PrecStd	120	25	25	30	20	35	38	67	51	39	31	24	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	11.6	16.9	23.7	26.1	29.5	31.7	31.8	26.9	21.2	14.8	10.6
		MCWB	7.9	7.7	10.5	14.9	17.7	21.4	20.2	20.4	18.0	15.3	11.6	8.4
	2%	DB	8.1	9.6	13.8	20.6	23.8	26.9	29.1	28.8	23.6	18.7	13.1	8.9
		MCWB	6.3	7.1	8.8	13.1	16.9	20.0	19.8	19.7	16.6	14.1	10.7	7.1
	5%	DB	6.3	7.9	11.5	17.9	21.8	24.9	26.9	26.6	21.5	16.6	11.7	7.1
		MCWB	4.7	5.7	7.7	11.5	15.6	18.5	19.0	18.8	15.7	13.2	9.4	5.6
	10%	DB	4.7	6.2	9.4	15.8	19.8	22.9	25.0	24.6	19.4	14.9	10.2	5.6
		MCWB	3.3	4.4	6.5	10.4	14.3	17.2	18.3	17.8	14.9	12.2	8.3	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	8.5	10.9	15.4	19.0	22.5	22.1	21.5	19.1	16.1	12.1	8.6
		MCDB	9.9	10.7	15.4	22.3	24.4	28.0	28.6	28.8	24.3	19.9	14.2	10.3
	2%	WB	6.4	7.2	9.3	13.7	17.3	20.5	20.6	20.5	17.5	14.7	10.9	7.2
		MCDB	7.9	9.3	13.0	19.5	22.5	25.9	26.9	27.1	22.4	17.8	13.0	8.7
	5%	WB	4.8	5.9	8.0	12.2	16.1	19.0	19.7	19.4	16.3	13.5	9.6	5.6
		MCDB	6.3	7.7	11.1	17.2	20.8	23.8	25.6	25.2	20.4	16.2	11.4	7.0
	10%	WB	3.4	4.4	6.7	10.8	14.9	17.7	18.7	18.5	15.3	12.4	8.3	4.4
		MCDB	4.5	6.1	9.2	15.0	18.9	21.9	23.8	23.5	18.8	14.7	10.0	5.5
Mean Daily Temperature Range		MDBR	4.9	5.8	7.4	10.0	10.4	10.4	10.9	11.4	9.3	7.4	5.0	4.5
	5% DB	MCDBR	5.5	7.8	12.0	14.3	14.3	14.5	15.4	15.2	13.5	10.6	7.0	5.2
		MCWBR	4.5	5.8	7.8	8.0	8.2	8.0	7.2	7.2	6.7	5.1	4.3	
		MCDBR	5.3	7.1	10.8	13.2	12.9	13.3	13.4	13.5	11.4	9.7	6.4	5.1
	5% WB	MCWBR	4.4	5.6	7.3	7.7	7.8	7.7	6.8	6.8	6.6	6.4	4.9	4.3
Clear-Sky Solar Irradiance	taub	0.305	0.331	0.374	0.423	0.421	0.418	0.437	0.429	0.392	0.365	0.338	0.302	
	taud	2.296	2.267	2.218	2.154	2.208	2.247	2.203	2.238	2.311	2.347	2.361	2.350	
	Ebn at Noon	716	791	818	816	837	841	816	802	793	743	672	666	
	Edh at Noon	68	91	115	137	136	132	135	125	104	84	65	57	
All-Sky Solar Radiation	RadAvg	0.83	1.55	2.60	4.10	4.99	5.26	5.00	4.49	3.17	1.86	0.91	0.62	
	RadStd	0.15	0.24	0.42	0.53	0.59	0.50	0.58	0.41	0.47	0.27	0.12	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	+0.78	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.48	N/A	N/A	N/A	+0.33	+0.30	N/A	N/A	+62	+24

Nomenclature: See separate page